Autosynth: Self-Calibrating Musical Instrument
Upon An Energy-Based Model

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Abstract

We present Autosynth, an interactive music instrument built on an energy-based model of the performer's own audio. Its control surface is not designed; it is measured from the model's own physics. The engine slices the user's tracks into beat-aligned fragments and, every bar, settles a free-energy minimization that decides which fragments play. Playing the instrument means pushing on that equilibrium and hearing it resettle. We ask what a knob on such a system can honestly be, and answer with a typed calculus: every legitimate control changes exactly one ingredient of the underlying problem—a cost, a temperature, a constraint, a frame of equivalence, or a clock—and each type behaves differently by necessity. The honest scalar knobs turn out to be the eigenmodes of the system's measured response at its current operating point: how many knobs exist, how strong each one is, and whether each is genuinely available are read off this measurement rather than chosen, and cross-talk between knobs vanishes in this basis. Three pre-registered measurements on the deployed system are reported, including a permutation test showing that a large apparent asymmetry between two controls was a fingerprint of the sequential sampler, not of the music. All negative results are retained.

Introduction

Generative models are typically *steered*: a human supplies a direction—a reward, a concept vector, a semantic setpoint—and machinery moves the model along it. This paper explores the converse construction: an instrument that derives its own control surface from the response physics of its trained equilibrium, so that the number of controls, their axes, their gains, their honest availability, their default limits, and even their display colors are measurements rather than design choices.

We build this in the domain of music. Autosynth arranges beat-sliced fragments of a user's own audio by settling a variational free energy every bar; playing it means tugging at a continuously moving equilibrium—the settled state wanders under its own dynamics—and hearing it re-settle around the pull. The central question is what a *control* on such an object can honestly be. Our answer is a typed calculus (see "A Typed Control Calculus"): every legitimate control perturbs exactly one datum of the variational problem and forms a conjugate pair with an observable the machine reports back; sensitivity is not designed but equals a constrained fluctuation (a fluctuation–dissipation identity at the operating point); and the honest scalar control basis is the eigenbasis of the response kernel, in which cross-talk vanishes and dead directions are absent rather than disabled.

The framework is falsifiable, and we report three instances in which it falsified its own deployment (see "Empirical Characterization"), including the retirement of a measured 45× response asymmetry as a sampler artifact via a pre-registered order-permutation test. We retain all negative results; several shaped the final interface.

Related Work

Inference-time steering. Steering a pre-trained generative model toward desired outputs is standard practice: exponential tilting of a base measure by a reward is the canonical formulation in inference-time alignment (Korbak, Perez, and Buckley 2022), with sequential

Monte Carlo and twisted variants for terminal or black-box rewards; classifier and classifier-free guidance play the analogous role for diffusion models (Ho and Salimans 2022). In language models, activation steering adds learned directions to hidden states (Turner et al. 2023), with recent control-theoretic treatments and multi-attribute extensions that confront interference between steering vectors, typically via imposed orthogonality constraints. All of these supply the steering *direction* from outside the model—a reward, a labeled contrast, a setpoint. Our controls differ in provenance: axes, gains, and availability are read off the model's own response kernel, and orthogonality is obtained rather than imposed.

Information geometry. Natural-gradient methods (Amari 1998) and exponential-family guidance exploit the Fisher geometry of a model class—the same conjugate structure (natural parameters and sufficient statistics) that underlies our tilt-type controls. We use this geometry at play time, as an interface, rather than at training time as an optimizer.

Response theory with generative models. A recent line combines the generalized fluctuation–dissipation theorem with score-based generative modeling to predict forced responses of stochastic systems from unperturbed statistics alone (Giorgini et al. 2024, 2025), including full parameter Jacobians as time-correlation integrals along a single baseline trajectory. Estimating response from unperturbed fluctuations—the move our joint-covariance measurement makes—is thus established machinery. Relatedly, the succession term $\mathrm{KL}(\pi \,\|\, K \circ \pi^*_{t-1})$ admits a Schrödinger-bridge reading (Léonard 2014; De Bortoli et al. 2021) (each bar as an entropic interpolation tethered to the propagated past), and bridge-side diagnostics such as reciprocal-consistency tests are a natural future check on the settlement stage; the casting stage departs from reciprocal structure by construction, which is exactly the departure our order-permutation test measures. Our contribution is narrower and interface-shaped: the *constrained, operating-point* projection of that identity (whose degenerate cases are the arming classes) and its use as the law that sizes, gains, and polices a performer-facing control surface.

Generation order in sequential models. That generation order matters in autoregressive models is well documented: fixed orders bias quality, any-order and learned-order models treat the ordering as a modeling choice, and order-averaging over sampled orderings has been used since NADE as an "orderless" inductive bias (Uria et al. 2016). Our fair (order-averaged) readout inherits that idea as an estimator. What we add is the use of order *permutation* as a pre-registered discriminator on a measured response asymmetry—collapse-to-floor and sign-flip under reordering, with cross-corpus constancy, classifying the asymmetry as sampler fingerprint rather than material property—deployed as an instrument diagnostic.

Latent-space instruments. The nearest neighbors are latent-space instruments in neural audio synthesis: RAVE exposes a variance-ordered, PCA-derived control space whose dimensionality is selected automatically from explained variance (Caillon and Esling 2021), and subsequent instruments in the NIME community reorganize such latents into performable dimensions, map gestures and tangible controllers onto latent axes, and treat latent spaces as first-class interaction surfaces. Self-sized control dimensionality read off a trained model is therefore established practice. The distinction here is which object is diagonalized: those instruments diagonalize a *representation* (the variance of a learned embedding—where the data sits), whereas Autosynth diagonalizes a *response* (the constrained covariance of an equilibrium at its operating point—how the settlement pushes back when forced). The distinction is empirically load-bearing: we exhibit a direction with ample representation-variance and zero response (see "Empirical Characterization"); a variance-ranked panel would ship that dead knob, a response-ranked panel cannot. Autosynth also contains no latent space at all—no encoder or decoder; its eigenpanel rotates the *force* basis of a variational problem over real audio slices. Elements with no cognate in the latent-instrument line include operating-point recomputation, arming as a degenerate case of a fluctuation–dissipation identity, an order-permutation discriminator separating object-level from sampler-level asymmetry, and frame-type drift meters with veto authority.

Honest overlap. Component-wise, nearly everything here is established: entropic transport and Schrödinger bridges; linear response and fluctuation–dissipation identities; exponential tilting; activation steering; natural-gradient guidance; PCA-derived self-sized control panels; Hodge and screw decompositions. What we believe is new is the conjunction and its direction—an instrument reading its control surface off its own response physics, with honesty (three-way arming) arising as a theorem's degenerate case rather than a design policy, limits from the material's own envelope, and names and colors from the spectrum. Each ingredient exists somewhere; we have not found the assembly.

The Engine

Autosynth ingests N music tracks; a beat clock slices beat-synchronous, band-split UNITS of real audio, provenance-tagged at birth and never stripped. Training freezes a WORLD: M self-sized anchors ("roles"), a succession kernel, contrastively-fit weights, and calibration statistics. M is the effective rank (participation ratio) of the cross-track traffic affinity, calibrated against a scrambled-geometry null—a *count of coupling rank*, not a guarantee of M pairwise-distinct roles: pruning removes only anchors with vanishing coupled mass, never duplicates, and degenerate mirror-pair anchors can occur (our committed demonstration world is exactly such a pair). At play, a causal WRITER settles each bar to the *mode* of a variational free energy given everything pinned—committed past, walls, user forces—then draws a finite-temperature realization around that mode and inks real slices to an output TAPE: the pipeline is entropic and stochastic end-to-end, with the minimizer appearing only as the center of the per-bar distribution, never as the emitted state. There is no decoder; rendering is choiceless stitching of the user's own audio (Fig. 1).

Formally,

$$\pi_t^*(u) \;=\; \arg\min_{\pi\in\Pi_t}\; F_t(\pi;u), \qquad F = \langle c,\pi\rangle + \varepsilon H(\pi) + \mathrm{KL}\big(\pi \,\big\|\, K\circ\pi_{t-1}^*\big) + V(\pi),$$

over a clamped polytope Π_t (committed past, boundary measures), quotiented by a gauge

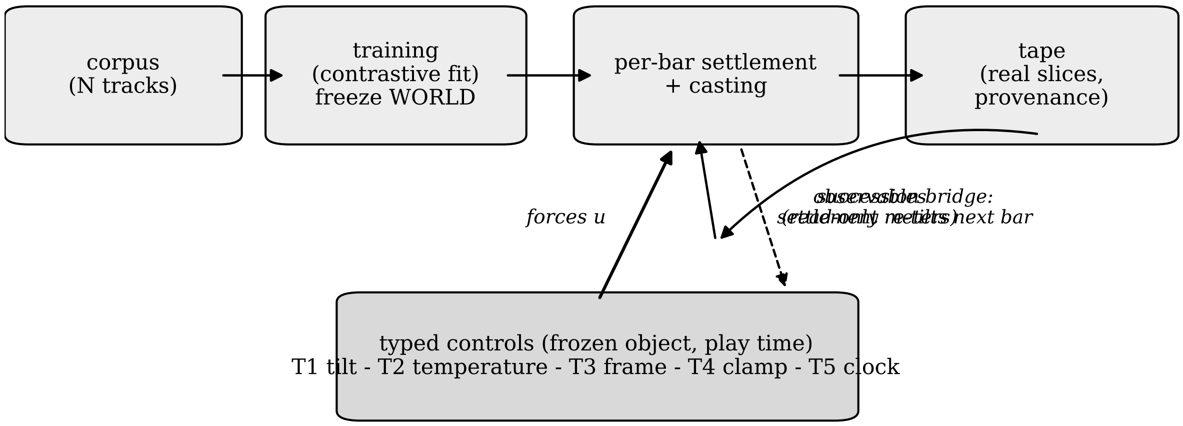

Figure 1: Training freezes the world; at play, typed controls (forces) enter the per-bar settlement, whose minimizer is inked to tape; observables return as read-only meters. Each settlement re-parameterizes the next bar's energy (forward propagation).

group G (frame symmetries); the entropic-transport machinery is standard (Cuturi 2013; Peyré and Cuturi 2019). Here K is the frozen succession kernel learned at training (a Markov operator on arrangements), and $K \circ \pi^*_{t-1}$ denotes its pushforward of the previous bar's settled state: the KL term is a bridge prior tethering each bar to the propagated image of its predecessor, which is how commitment propagates forward without any bar revising the past. Note F is entropy-regularized (εH), so π^* is the *mode* of the induced Gibbs measure—a smoothed optimum, not a hard combinatorial argmin—and the emitted bar is a finite-temperature draw around π^* (settlement centers the distribution; sampling realizes it). For $\varepsilon > 0$ and under standard regularity at the solution (LICQ, strict complementarity, and second-order sufficiency on the active face—conditions the entropic term makes generic in the interior and which we assume, not verify, at face boundaries), the minimizer is unique and C^1 in u by the implicit function theorem on the KKT system; when the active set changes across bars the response kernel is only piecewise smooth, and all operating-point statements below are local to the current face; all response theory below is differentiation of that system. F is a variational free energy; the induced per-bar distribution is a Gibbs measure. Training solves the inverse problem—corpus bars are minima of their scramble orbits under a noise-contrastive fit (Gutmann and Hyvärinen 2010)—after which parameters freeze. Controls act at play time on the frozen object.

The two-stage realization: certified settlement, then sequential casting. Each bar is produced in two stages that steer orthogonal quantities. SETTLEMENT decides *how much of each role* the bar holds: a block-coordinate entropic-mirror descent on the bar's occupancy block $O \in \mathbb{R}^{M \times S}$ in the field of the frozen anchors, accepted only under a Lyapunov descent certificate—an uncertified bar is never emitted. This stage is a genuine *joint* equilibrium solve: the exact object the response theorems describe. CASTING then fills each role-stamped slot with *source material*: a temperature draw around the settled mode, then a sequential, slot-by-slot fiber sweep in which each candidate is scored by F's own energies plus the play-time tilts and drawn by Gumbel-max (Maddison, Tarlow, and Minka 2014)—i.e. exact categorical *sampling* from the temperature-T_s softmax, not argmax selection; the stochasticity is the point.

Design choice: the casting sweep is sequential (greedy), deliberately. The sweep is causal—it plays through the bar and cannot revise committed slots—the only shape a live performer can have; a simultaneous casting is a composer's-eye object, outside time. The choice is also empirically forced: a moment-matched simultaneous variant produced markedly lifeless output—taste without commitment. The sweep's irreversible slot-ordering is retained as the instrument's identity, and it is precisely localized: the performer's arrow lives in *casting*; settlement remains the certified joint solve. The sweep breaks detailed balance, so the deployed response on casting-stage lanes is that of a causal process rather than of the equilibrium it targets—settled *states* still answer to F, while the casting *response* bears the performer's arrow (see "Empirical Characterization"); that signature is characterized and subtracted in measurement (order-averaged readouts) and kept in performance.

Where the control lanes enter. The stage split types the tilt lanes: *region* and *density* act on the occupancy block at settlement; *continuity* and *novelty* act on the casting logits, $\log w(c) = -E_F(c)/T_s + \lambda_{\text{cont}} \mathbf{1}[\text{cont}](c) + \lambda_{\text{nov}}\, \text{reuse}(c) + \beta(c)$, with β the material-lean addend. All control reaches the writer through a single tilt carrier with one construction point (anything else is a type error), and a neutral carrier is bit-identical to no carrier. One structural consequence, predicted by conjugacy and confirmed in code, deserves note: a *pure-role* addend at casting is provably inert—within one fiber choice set every candidate shares the settled role, so a role-only weight is a softmax constant that cancels in the Gumbel-argmax. Role *amount* is therefore steerable only at settlement: a force that does not vary across the choice set has no conjugate observable and is no control. The wall is not policy; it is the calculus. The same structure explains why material-lean (track-level) steering is well-typed at casting even when the anchor-by-band profile is degenerate: the casting addend keys on the candidate's source identity directly and never routes through the profile, so it varies across the choice set regardless of the profile's rank.

Two motions are intrinsic: *forward progression* (each settlement re-parameterizes the next bar's energy through the succession bridge) and *breathing* (the equilibrium is a soft basin; per-bar sampling within it yields living variation). The instrument is publicly playable at `autosynth.fun`, so reviewers can play what this article describes; the site carries no author-identifying information. In deployment, cloud training operates on gauge-invariant stage-3 cost matrices only, so that in local sealed mode the gauge boundary coincides with the privacy boundary; trained worlds are shared without raw audio, with device-verifiable receipts.

A Typed Control Calculus

Before the formal statement, the intuition. A settled bar is the answer to a question: "given this cost, this much softness, these fences, this notion of equivalence, and this playback map, which arrangement wins?" Written down, that question has exactly five ingredients—the cost c, the temperature ε, the feasible set Π, the gauge group G, and the output map—and nothing else. So if a knob is going to change the answer honestly, it must change one of those five ingredients, and which ingredient it changes determines everything about how it behaves: what it feels like to move, what the machine can report back about it, and what can go wrong with it. A control that changes the *cost* is a lean: it makes some arrangements cheaper, the equilibrium shifts toward them, and the machine can report exactly how far it actually moved—the conjugate observable. A control that changes the *temperature* is not a direction at all but a loosening of everything at once, so it has no target to report, only a spread. A control that moves a *fence* does not lean; it forbids, and the system either presses against it or ignores it. A control acting through the *gauge*—reframing what counts as the same arrangement—is the only one for which the route taken matters, so it alone carries path memory. And a control on the *output map* changes playback without changing any decision. This is the entire content of the typing: five ingredients in the question, therefore five kinds of knob, each with its own signature—and anything sold as a sixth kind must secretly be one of these or a fabrication.

Controls are not parameters. Trained parameters are frozen and their formal conjugates are not runtime percepts; a legitimate control perturbs exactly one datum of $(c, \varepsilon, \Pi, G, \text{output map})$: A *tilt* (T1) shifts the cost, $c \mapsto c - u\,\varphi$, forming a conjugate pair between the force u and the observable $\Phi = \langle \varphi, \pi \rangle$. A *temperature* control (T2) scales ε; it is directionless and reshapes every equilibrium at once. A *frame* control (T3) acts through the gauge group G; its invariant response splits into a symmetric displacement (slide) and an antisymmetric cycle residue (loop), the unique carrier of path memory (von Neumann and Schoenberg 1941; Masani 1972). A *clamp* (T4) moves the feasible polytope Π; it binds rather than leans, and it is the one type whose content is definitionally external, since a wall is where a human fences the space. A *clock* (T5) post-composes the output map and commutes with the optimization, so the settled schedule is invariant under tempo. The classification is read off the data of the variational problem, not imposed on it: these are the five things the problem *has*, so they are the five things a control can touch. Each type carries a distinct action geometry (a signed bias with a conjugate readout; a set-and-stays scale; a wall; a path with memory; a clock), of which the deployed interface currently exposes the first two. One epistemic point applies to all of them: control is never observed on the Gibbs object directly. Settlement-stage tilts move the *mode* of the per-bar measure and casting-stage tilts reweight the *draw*, but in both cases what the performer hears—and what every meter reads—is the finite-temperature realization. All steering is seen through the sampler; the equilibrium object is inferred, never inspected.

Theorem 1 (Operating-point fluctuation–dissipation)**.** $\dfrac{\partial\langle\Phi\rangle}{\partial u} = \dfrac{1}{\varepsilon}\,\mathrm{Cov}_{\pi^*}(\Phi,\Phi)$, *projected onto the tangent of the active constraints at the current state.*

The projection is essential: an observable saturated at a polytope face has across-bar drift variance but approximately zero constrained fluctuation—an *armed-but-inert* control. Drift-variance calibrations are therefore state-mismatched estimators; the correct calibration is the constrained covariance at the operating point.

Corollary 1 (Three-way arming)**.** *A lane is* STEERABLE *(measured $\sigma_\varphi > 0$: the force applies a real tilt $\lambda = u/\sigma$),* DEGENERATE *(identifiable yet $\sigma_\varphi = 0$ exactly: the lane resolves to the identity tilt, a proven-constant direction), or* DISARMED *(not identifiable: no valid scale exists; no tilt is invented, though the force is still transmitted and recorded). The classes are computed from untilted settlement statistics with no invented floor. All three are cases of Theorem 1; none is policy.*

Theorem 2 (Reciprocity, scoped)**.** *For the exact joint-Gibbs object, tilt–tilt cross-talk is symmetric: $\partial\langle\Phi_A\rangle/\partial u_B = \frac{1}{\varepsilon}\mathrm{Cov}(\Phi_A,\Phi_B) = \partial\langle\Phi_B\rangle/\partial u_A$. A deployed sequential sampler is not that object; its kernel need not be symmetric, and order-dependent asymmetries are algorithm fingerprints. Discriminator: permute the sampler's processing order with the object untouched—object-level asymmetry survives; artifacts collapse or flip.*

Theorem 3 (Gap and release)**.** *The target–achieved gap is governed by the same kernel; on release the settlement relaxes along the unforced equilibrium path. Displayed gaps, release trajectories, and response magnitudes must derive from engine state; any interface easing or shaping of them misrepresents the system.*

The full response kernel decomposes as a symmetric tilt/covariance block plus an antisymmetric frame part; informational content that survives gauge and symmetrization lives in the second summand. On a sequential sampler, however, the measured antisymmetric part is contaminated by ordering bias; only an order-averaged (fair) readout's residual counts as object-level.

The Self-Calibrating Interface

On the deployed surface, the control structure—which knobs exist, along which axes, with what gains, and whether they are honestly available—is measured, not chosen. The calculus and the measurement thus divide the work between them. The five types are read off the *form* of the variational problem, before any training: they say what kinds of control can exist and how each kind must behave. The operating-point measurement is read off the *content* of the trained object: it says which controls actually exist on this corpus, how many clear the noise floor, how strong each is, and whether each is honestly available. The typing also tells the measurement what to measure and what its readouts mean—sensitivity is a conjugate fluctuation because the tilt type says so; temperature reports a spread because its type has no direction. Nothing in either layer is designed: one is fixed by the logic of the problem, the other by the corpus.

Spectral Controls

Diagonalizing the constrained covariance at the operating point yields the scalar control basis: knobs are eigenmodes above the measured noise floor; their count k self-sizes by the same effective-rank law that sets M; gains equal eigenvalues, so calibration holds by construction and the drift-variance mismatch of Theorem 1 dissolves; cross-talk vanishes in the eigenbasis, trivializing Theorem 2. Every arming pathology of an earlier hand-named lane family (degenerate lanes, an inert lane, asymmetric-looking cross-talk) was the cost of reading a non-diagonal kernel through a basis the object does not use (Fig. 2).

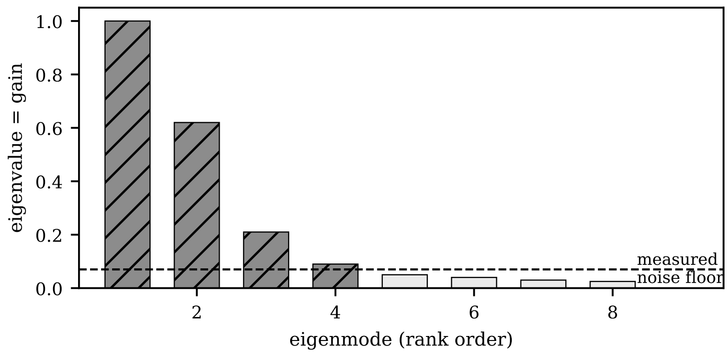

Figure 2: The panel self-sizes: knobs are eigenmodes of the constrained covariance above the measured noise floor. Eigenvalues are the gains; modes below the floor are absent.

Estimation is documented for scrutiny: the covariance is estimated on the sampling ensemble at the current operating point over a sliding window of settled bars at the current T_s; the noise floor is the same statistic computed under a scrambled null (independently permuted geometry, destroying shared structure), pre-registered as the single cutoff rule used at every scale; eigenmodes are recomputed on a cadence rather than continuously. We report point estimates without bootstrapped confidence intervals in this version, and we flag mode stability across time and temperature as an explicitly open measurement (see "Limitations and Open Questions"): modes near the floor can enter and leave the panel, and the panel is honest about the current window, not about permanence.

As deployed, each mode is rendered as a lean strip carrying two marks: the performer's handle (the force) and a machine mark showing the achieved projection of the settled state, read from live telemetry. The gap between them is the settlement resisting (Theorem 3); on release the force is zeroed and the machine mark simply continues to report the unforced state—no interface animation writes either mark. Modes are operating-point objects, recomputed on a cadence, and directions below the floor are absent rather than disabled. The retired hand-named lanes remain available in a collapsed legacy drawer, honestly classed (steerable/degenerate/disarmed) per corpus.

The Material Surface

Material selection is a separate, spatial surface. As deployed it is a track$\times$role grid whose geometry mirrors the two stages: *columns* (roles) steer settlement through the region lane; *rows* (tracks) and *cells* (track, role) steer casting through the material-lean addend; a disarmed or degenerate lane is simply absent from the emitted payload. The two axes of the grid have different provenances, and the difference is part of the design ethic. Tracks are *given*: they are the identities of the material the performer loaded, carried as provenance from ingestion, learned by nothing—and track names are therefore the one place user vocabulary legitimately appears on the surface, since they name the performer's own files rather than the model's structure. Roles are *measured*: the corpus's own sound categories, their count self-sized against the noise floor. The surface thus crosses one given axis with one measured axis; together with the logically fixed types, everything on the instrument traces to one of three provenances—given by the material, measured from the object, or forced by the form of the problem—and none of the three is designed. Interaction is a single signed gesture: scroll or drag up to amplify, down to de-amplify, at a row, a column, or a cell—soft and saturating, never a hard mute (hard membership would be a clamp, a different type, not currently exposed). The read side is strictly separated from the write side: every glow is a read-only reduction of produced placements or committed occupancy, calling nothing downstream, so audio is byte-identical whether or not the display runs; and the committed tape records placed units only, never the bias—steering leaves a transient wake in the working tape, not a written weight. The governing display law: nothing reaches a pixel without passing through F. Its two observable signatures—co-movement of untouched elements and state-dependent response to identical pushes—are emergent from re-settlement and cannot be produced by a display that echoes input.

Empirical Characterization

Three pre-registered, read-only measurements on the deployed system are reported here, in the order they were run. Each changed the deployment; none was tuned after the fact.

A uniform coupling made profile-routed levels degenerate—and the durable fix was structural. The anchor-by-band profile was uniform on early trained worlds: a track-level bias routed *through the profile* collapsed onto the global density marginal—a lean whose label lied—and profile-based unit grouping collapsed with it. Both were disarmed; role-level tilt, which does not route through the profile, remained live. The lasting resolution came from the stage split itself: material-lean steering was re-typed onto the *casting* stage, whose addend keys on candidate source identity and never touches the profile—track and $(\text{track}, \text{role})$ steering are thereby well-typed regardless of the profile's rank, and only the profile-routed *displays* (unit grouping; per-track role decomposition) await the pre-registered profile change. When unit grouping returns it must render as soft role-mass ranking, since the coupling is a soft assignment and a partition tree would fabricate hierarchy.

An armed lane was inert. The continuity lane, armed by its drift-variance calibration, was mechanically inert: the corpus sits approximately 91% saturated on that observable. This finding forced Theorem 1's operating-point statement and the three-way classification; the shipped rendering shows the condition honestly (the handle drags, the machine mark stays, an at-ceiling note surfaces from live telemetry only).

A $45\times$ asymmetry was an artifact. Novelty appeared to suppress continuity roughly $45\times$ harder than the reverse—a prima facie violation of Theorem 2. We define the cross-talk asymmetry $dK := \hat{K}_{\text{nc}} - \hat{K}_{\text{cn}}$, where $\hat{K}_{ij}$ is the finite-difference estimate of $\partial\langle\Phi_i\rangle/\partial u_j$ between the novelty and continuity lanes; the *coordinate correction* divides each entry by the product of the lanes' measured scales $(\sigma_{\varphi_i}\sigma_{\varphi_j})$, so dK is dimensionless and invariant to per-lane units—the raw $45\times$ ratio is a unit-dependent restatement of the same measurement. The pre-registered order-permutation discriminator collapsed dK from -0.417 (default order) to

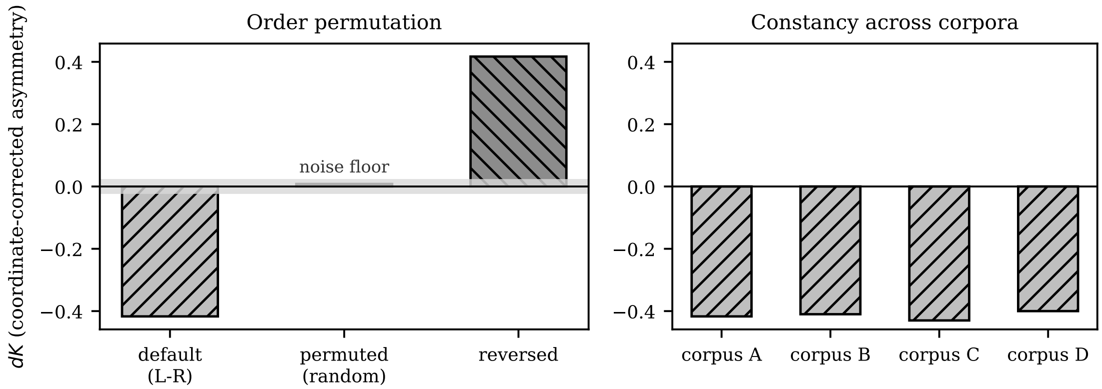

Figure 3: Left: the coordinate-corrected asymmetry dK collapses to the noise floor under random slot-order permutation and flips sign under reversal—impossible for a material asymmetry. Right: dK is approximately constant across corpora—an algorithm fingerprint.

+0.008 (the noise floor) under random slot-order permutation, flipped its sign under order reversal, and found it approximately constant across four corpora (Fig. 3): the asymmetry is the casting sweep's slot-ordering fingerprint, not the music's directedness. The localization is exact: continuity and novelty are casting-stage lanes, so their cross-talk is measured through the sequential sweep; region, a settlement-stage lane, exhibited no such anomaly—consistent with settlement being the certified joint solve. Mechanistically, the default order builds near-ceiling continuity runs, leaving novelty maximal runs to break. Reciprocity is thereby rescued at its correct scope; an earlier reading of the asymmetry as intrinsic non-reciprocal content is retired. The sampler itself is retained unchanged—its output is the instrument's identity—and remediation is measurement-side: the ordering bias is characterized and subtracted, and both raw and order-averaged readouts are reported. Whether any object-level cycle residue survives fair measurement remains open.

Limitations and Open Questions

The residual cycle-residue question is open: on the deployed sampler the antisymmetric response is dominated by the characterized ordering bias, and whether any object-level holonomy survives order-averaged measurement is undetermined at our fixture scale. The uniform-coupling degeneracy limits the interface to two honest levels until the pre-registered engine change lands. Eigenmodes are operating-point objects and rotate with the state; whether performers experience this as expressive or disorienting is an empirical HCI question we have not studied. The one methodological convention throughout is the noise-floor cutoff rule, pre-registered once and applied identically at every scale. The mode measurement's observable scope requires care: if the response kernel is estimated on settlement-stage observables alone, casting-stage structure is invisible to it and the reported mode count is a lower bound on the instrument's steerable directions—our current mode-count diagnosis treats this explicitly, alongside temperature-dependence (modes freeze in and out with the sampling temperature, so the panel is the temperature's-eye view of the object) and operating-point saturation. One specified control lane (a second-moment/anisotropy tilt present in the carrier) remains dormant and unsurfaced pending typing. This version reports point estimates without confidence intervals, no controlled comparison against latent-variance instruments on dead-knob rate or interference, and no listening or user studies; the in-flight measurement program (joint-covariance re-estimation, temperature-sliced spectra, order-averaged cross-talk with residual-asymmetry bounds) is designed to supply the quantitative half of that list, and the comparative and human-subject half is future work we consider necessary before strong practical claims. Finally, the novelty assessment in the related-work discussion is our own reading of the closest lines; a systematic occupancy review may reassign parts of the conjunction.

Conclusion

Autosynth demonstrates a generative instrument whose control surface is measured: knob count, axes, gains, and honest availability are read off the trained object, with honesty arising as the degenerate case of a fluctuation–dissipation identity rather than as policy. The same measurements that calibrate the surface also police it: a mislabeled control was disarmed, an inert one was exposed before it shipped, and a measured asymmetry was retired as a sampler artifact through a pre-registered discriminator—while the musical output, the instrument's identity, was left untouched throughout.

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